Lab 2 函式應用、異質與結構矩陣

函式建立

sample mean: $\mu_x = \dfrac{\sum_{k=1}^{N} x_k}{N}$

```
X=[3 1 8 6 5 4 3 5 4 1 6 8 7 9 6];
N=length(X);
mu_x=sum(X)/N
```

```
mu_x = 5.0667
```

sample variance: $S_x = \dfrac{\sum_{k=1}^{N} (x_k - \mu_x)^2}{N-1}$

```
S_x=sum((X-mean(X)).^2)/(N-1)
```

```
S_x = 5.9238
```

Build up a function "mv.m" to calculate sample mean and variance

```
[mu, S]=mv(X)
```

```
mu = 5.0667
S = 5.9238
```

function規則

(1) start with "function"

(2) function name = file name

(3) input: right-hand side, with parenthesis (optional)

(4) output: left-hand side, with bracket (optinal)

(5) ; %換行符號/不輸出結果至**command window**

(6) = % 指派等號右邊數值給左邊變數

function v.s. workspace

Variables created in functions cannot be called outside the function.

Transport Variables

(1) Call by address: sent the memory address

(always deal with one variable)

```
global a b
a=10; b=20;
[c d]=cba
```

```
c = 30
d = -10
```

(2) Call by value: copy value

(deal with several identical variables)

```
e=10; f=20;
[g, h]=cbv(e,f)
```

```
g = 30
h = -10
```

Global Variables

MATLAB also provides call-by-address approach.

You have to declare global variables before using them.

Flexibility in Using Functions

(1) Various numbers of inputs

(2) Setup defualt values of variables

(3) Print out the customized error message

Input/output Flexibility

(1) nargin: number of function input arguments.

```
e=10;
[g, h]=cbv(e)
```

```
ans = 1
g = NaN
h = NaN
```

(2) nargout: number of function output arguments.

(3) varargin: variable length input argument list.

(4) varargout: variable length output argument list.

異質矩陣 (cell array)

(1) Store different data formats in a single cell array.

(2) Number, strings, arrays

Archive Big Dataset

For example,

- 30 normal controls vs. 30 patients with stroke

- Name, ID → string
- Age, gender, height, weight → numbers
- Pre-training data (8 x 450 values) → number array
- Post-training data (8 x 450 values) → number array

```
load('ExampleCell.mat')
```

Create a Cell array

```
data={}; % create an empty cell array
```

```
data=cell(4,1); % create a cell array
data(1)={'NTUST'}; % string
data(2)={47}; % number
data(3)={rand(10,100)}; % number array
data(4)={{100, 'no. 1 technology university'}}; % number array
```

```
data=cell(4,1); % create a cell array
data{1}='NTUST'; % string
data{2}=47; % number
data{3}=rand(10,100); % number array
data{4}={100, 'no. 1 technology university'}; % number array
```

```
List1=['David ';'Andy  ';'Jay   ';'Jolin ';'Selina'];
List2={'David';'Andy';'Jay';'Jolin';'Selina'};
List2(1)
```

```
ans = 1×1 cell array
    {'David'}
```

```
List2{1}
```

```
ans =
'David'
```

Useful functions

(1) cell: create cell array

(2) iscell: true for cell array

```
iscell(List2(1))
```

```
ans = logical
   1
```

```
iscell(List2{1})
```

```
ans = logical
   0
```

Index of a cell array

```
A={rand(6,4), rand(5,7)};
A{1,2}(4,5)
```

```
ans = 0.5051
```

```
A{1,1}(2:4,2:3)+A{1,2}(2:4,5:6)
```

```
ans = 3×2
```

```
0.5855    1.0426
1.0396    1.1730
0.5758    0.4688
```

Archive Big Dataset

For example,

- 30 normal controls vs. 30 patients with stroke

- Name, ID → string
- Age, gender, height, weight → numbers
- Pre-training data (8 x 450 values) → number array
- Post-training data (8 x 450 values) → number array

```
load('ExampleStruct.mat')
```

結構矩陣 (structure array)

(1) Store different data formats in a single structure array.

(2) Number, strings, arrays

NC

PLOTS | VARIABLE | VIEW

New from Selection | Open | Print | Rows | Columns | Insert | Delete | Transpose | Sort

VARIABLE | SELECTION | EDIT

1x30 struct with 7 fields

Fields	name	age	gender	height	weight	predata	postdata
1	'NC01'	67	1	168.9951	79.9885	8x450 do...	8x450 dou...
2	'NC02'	59	1	170.9523	63.6980	8x450 do...	8x450 dou...
3	'NC03'	57	1	161.8978	52.8100	8x450 do...	8x450 dou...
4	'NC04'	63	1	168.9596	57.9199	8x450 do...	8x450 dou...
5	'NC05'	68	0	167.8267	62.2405	8x450 do...	8x450 dou...
6	'NC06'	67	1	169.2874	68.6386	8x450 do...	8x450 dou...
7	'NC07'	57	1	173.4562	59.3461	8x450 do...	8x450 dou...
8	'NC08'	65	1	161.4962	51.5521	8x450 do...	8x450 dou...
9	'NC09'	63	1	163.6110	51.5550	8x450 do...	8x450 dou...
10	'NC10'	63	1	167.6989	73.5794	8x450 do...	8x450 dou...
11	'NC11'	61	0	160.4540	65.5484	8x450 do...	8x450 dou...
12	'NC12'	60	1	162.0846	68.6749	8x450 do...	8x450 dou...
13	'NC13'	59	0	165.8496	68.3210	8x450 do...	8x450 dou...
14	'NC14'	50	0	164.8730	53.5264	8x450 do...	8x450 dou...
15	'NC15'	52	1	172.6919	57.2291	8x450 do...	8x450 dou...
16	'NC16'	53	0	163.8809	59.1720	8x450 do...	8x450 dou...
17	'NC17'	64	1	168.6986	73.7783	8x450 do...	8x450 dou...
18	'NC18'	62	1	164.8790	70.9246	8x450 do...	8x450 dou...
19	'NC19'	52	0	171.6970	68.7181	8x450 do...	8x450 dou...
20	'NC20'	56	0	164.9102	53.6330	8x450 do...	8x450 dou...
21	'NC21'	55	1	166.5719	63.4037	8x450 do...	8x450 dou...
22	'NC22'	59	1	171.1880	67.7463	8x450 do...	8x450 dou...
23	'NC23'	66	0	160.9877	70.1898	8x450 do...	8x450 dou...
24	'NC24'	55	0	170.2819	73.0139	8x450 do...	8x450 dou...
25	'NC25'	60	1	170.8547	76.9674	8x450 do...	8x450 dou...
26	'NC26'	62	0	166.0561	51.9179	8x450 do...	8x450 dou...
27	'NC27'	51	0	162.1981	78.4609	8x450 do...	8x450 dou...
28	'NC28'	56	1	168.7833	53.3143	8x450 do...	8x450 dou...
29	'NC29'	55	1	166.1138	56.4616	8x450 do...	8x450 dou...
30	'NC30'	63	0	171.1190	71.4411	8x450 do...	8x450 dou...
31							

NC

PLOTS | VARIABLE | VIEW

New from Selection | Open | Print | Rows 1 | Columns 1 | Insert | Delete | Transpose | Sort

VARIABLE | SELECTION | EDIT

31x7 cell

	1	2	3	4	5	6	7	8
1	'name'	'age'	'gender'	'height'	'weight'	'predata'	'postdata'	
2	'NC01'	62	1	172.1076	69.4049	8x450 do...	8x450 do...	
3	'NC02'	57	0	170.2542	52.2071	8x450 do...	8x450 do...	
4	'NC03'	58	1	174.2668	72.6109	8x450 do...	8x450 do...	
5	'NC04'	63	1	173.4476	69.0174	8x450 do...	8x450 do...	
6	'NC05'	65	1	168.8933	58.0383	8x450 do...	8x450 do...	
7	'NC06'	52	1	168.6337	58.7737	8x450 do...	8x450 do...	
8	'NC07'	65	0	165.8925	79.0742	8x450 do...	8x450 do...	
9	'NC08'	54	0	169.7196	79.6230	8x450 do...	8x450 do...	
10	'NC09'	52	0	174.6123	61.6098	8x450 do...	8x450 do...	
11	'NC10'	68	1	162.9426	64.0754	8x450 do...	8x450 do...	
12	'NC11'	68	1	163.8363	69.6155	8x450 do...	8x450 do...	
13	'NC12'	63	1	172.5047	79.3612	8x450 do...	8x450 do...	
14	'NC13'	55	1	173.8082	77.4354	8x450 do...	8x450 do...	
15	'NC14'	51	0	169.6144	77.6480	8x450 do...	8x450 do...	
16	'NC15'	54	1	169.6435	67.4950	8x450 do...	8x450 do...	
17	'NC16'	59	1	168.1967	65.7889	8x450 do...	8x450 do...	
18	'NC17'	62	1	170.1513	51.4973	8x450 do...	8x450 do...	
19	'NC18'	50	0	164.3884	56.8752	8x450 do...	8x450 do...	
20	'NC19'	53	0	161.9704	59.7239	8x450 do...	8x450 do...	
21	'NC20'	68	0	163.0854	75.9676	8x450 do...	8x450 do...	
22	'NC21'	63	1	161.4267	66.9368	8x450 do...	8x450 do...	
23	'NC22'	63	0	171.5488	67.1798	8x450 do...	8x450 do...	
24	'NC23'	58	1	163.9583	78.3704	8x450 do...	8x450 do...	
25	'NC24'	51	0	168.6894	72.8220	8x450 do...	8x450 do...	
26	'NC25'	66	0	162.9342	79.3600	8x450 do...	8x450 do...	
27	'NC26'	62	1	165.3817	66.6387	8x450 do...	8x450 do...	
28	'NC27'	52	1	173.2677	69.8264	8x450 do...	8x450 do...	
29	'NC28'	51	1	163.8164	73.5234	8x450 do...	8x450 do...	
30	'NC29'	53	1	160.2950	59.9052	8x450 do...	8x450 do...	
31	'NC30'	68	1	167.7351	72.8175	8x450 do...	8x450 do...	

Create a Structure array

```
data.name='NTUST'; % string
data.age=47; % number
data.predata=rand(10,100); % number array
data.notaion={100, 'no.1 technology university'}; % cell array
```

```
data(2).name='NTUST'; % string
data(2).age=47; % number
data(2).predata=rand(10,100); % number array
data(2).notaion={100, 'no.1 technology university'}; % cell array
```

```
List1=['David ';'Andy  ';'Jay   ';'Jolin ';'Selina'];
List2(1).name='David';
List2(2).name='Andy';
List2(3).name='Jay';
List2(4).name='Jolin';
List2(5).name='Selina';
```

```
data(1).predata(4,3)
```

```
ans = 0.2336
```

```
data(1).predata(2:5,3:4).*data(1).predata(3:6,5:6)
```

```
ans = 4×2
    0.0929    0.0940
    0.1382    0.2368
    0.1255    0.0131
    0.0066    0.0303
```

Useful functions

(1) struct: create or convert to structure array

```
data=struct('name','NTUST', 'age', 47)
```

```
data = struct with fields:
    name: 'NTUST'
     age: 47
```

(2) isstruct: true for structures

```
isstruct(data)
```

```
ans = logical
   1
```

(3) fields: display a list of fields in a structure array

```
fields(data)
```

```
ans = 2×1 cell
'name'
```

```
'age'
```

(4) isfield: true if field is in structure array

(5) rmfield: remove fields from a structure array

Benefits for using a single array

(1) easy to clear, save, and load

(2) easy to make it a global variable

(3) easy to categorize variables

```
function [mu, S]=mv(X)
N=length(X);
mu=sum(X)/N;
S=sum((X-mean(X)).^2)/(N-1);
end

function [c, d]=cba
% call-by-address demonstration
global a b
c=a+b;
d=a-b;
end

function [sum, diff]=cbv(num1, num2)
% call-by-value demonstration
  if nargin<2
     nargin
     sum=nan;
     diff=nan;
     return
  end

sum=num1+num2;
diff=num1-num2;
end
```